Planner vs solver on generated Kleine Binckhorst scenarios

20 scenarios per cell, 5 train counts × 3 matchings = 300 scenarios; planner (symbolic-rail, image edge) and solver run on identical instances; generator, solver and evaluator `stable/stable/stable`; no servicing; **300 s per scenario for both tools**. A scenario counts as solved only if the TORS evaluator accepts the plan.

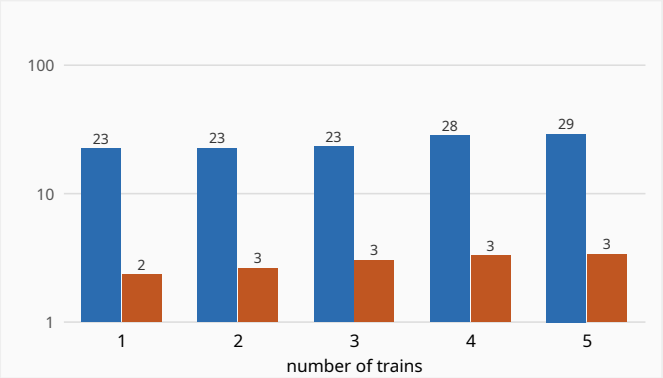
1. How many scenarios are solved?

Overall the two tools are level (80% planner, 79% solver). The planner is clearly ahead on random matching; the solver is ahead on FIFO and LIFO.

trains	FIFO		LIFO		random	
	planner	solver	planner	solver	planner	solver
1	100% (20/20)	100% (20/20)	100% (20/20)	100% (20/20)	100% (20/20)	65% (13/20)
2	95% (19/20)	100% (20/20)	95% (19/20)	100% (20/20)	95% (19/20)	65% (13/20)
3	90% (18/20)	100% (20/20)	100% (20/20)	100% (20/20)	55% (11/20)	35% (7/20)
4	80% (16/20)	100% (20/20)	80% (16/20)	95% (19/20)	35% (7/20)	15% (3/20)
5	70% (14/20)	85% (17/20)	60% (12/20)	90% (18/20)	40% (8/20)	30% (6/20)
all	**87%** (87/100)	**97%** (97/100)	**87%** (87/100)	**97%** (97/100)	**65%** (65/100)	**42%** (42/100)

2. Time to a plan

The solver needs about 2–3 s; the planner needs about 23–29 s.

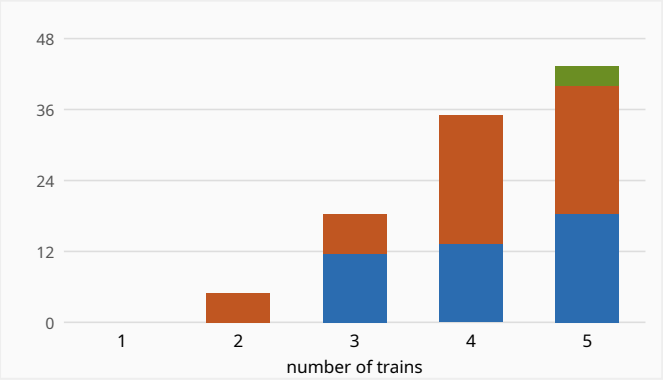


planner solver

y-axis: median seconds to an accepted plan (log scale). Planner times for 1–3 trains are the median over 5 repetitions of those scenarios; other train counts come from the single sweep.

3. Why the planner fails

Failures start at 2–3 trains and are dominated by late departures and search timeouts.



search timeout late/early departure track conflict other

y-axis: % of the planner's scenarios at that train count (60 each) that fail, by cause.